

Validation through Replication of Augmented Reality as a Visualization Technique for Scholarly Publications in Astronomy

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Experimental Design of Conceptual Replication

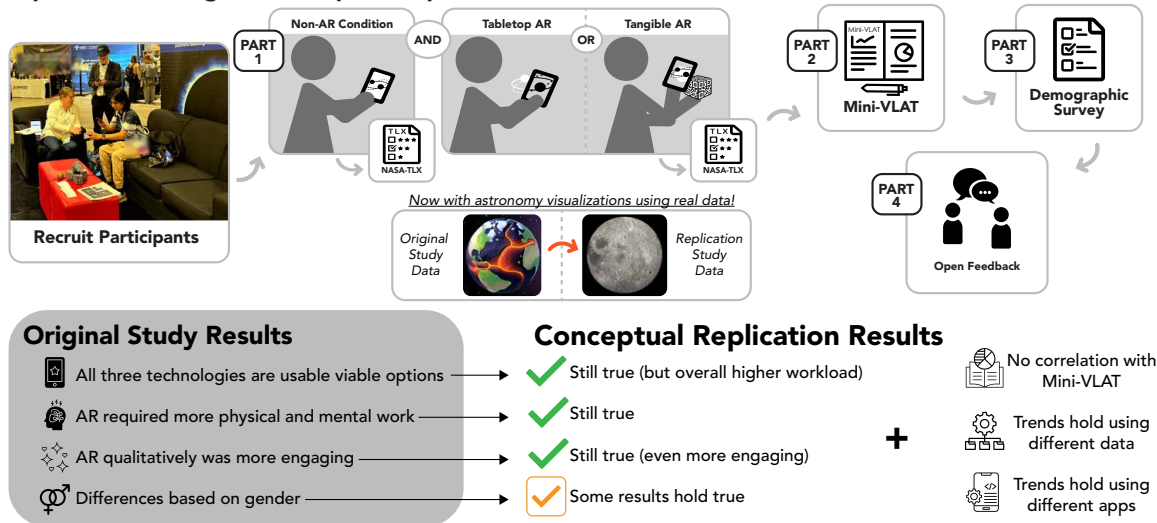


Figure 1: The procedures and results of a conceptual replication of a prior study conducted by Adams et al. [1]. We highlight one example of the difference between stimuli in the studies, which now use real data, and summarize our findings which mostly verify the results of the original study, strengthening its validity.

ABSTRACT

The fields of astronomy and astrophysics are evolving and incorporating technologies to effectively view and explore 3D data visualizations, including Augmented Reality (AR). An analysis of the feasibility of using AR in journal publications for 3D visualizations took place two years ago, in 2023, when Adams et al. evaluated whether the perceived workload between AR and non-AR technologies was comparable. Given that the use of AR for astronomy journal publications was, and still is, in its infancy, the original study had to utilize data intended for K-12 education that had similar interactions and data types as a proxy for real-world data that could be visualized in future astronomy publications. In this paper, we present the results of a conceptual replication study of Adams et al.'s work to validate whether their findings hold with real astronomy stimuli. We found in our replication that many of the trends in the original study hold true, but that the workload experienced by participants was significantly higher under multiple conditions when using real-world data. Additionally, we found that the tradeoff between engagement and workload was as prevalent in the replication as it was in the original study. Our results provide a new framing for researchers to understand the tradeoffs of immersive visualization technologies and the increased workload of pairing these tools with complex, scientific stimuli. All Supplemental Material in our study is available at <https://osf.io/j8urq/>.

Index Terms: Augmented Reality, Human Subjects - Empirical Study, Astronomy Visualization, Replication Study, AR-Enhanced Publications

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1 INTRODUCTION

Astronomy and astrophysics produce a plethora of multidimensional data that captures the beauty and magnitude of the universe, however, conventional static visualization methods fail to effectively represent this data. To represent more of the data at once, 3D visualization mediums such as scatterplots, isosurfaces, and volume renderings are used [21]. Despite the 3D nature of the visualizations being generated, there is still a disconnect with using 2D modalities, such as laptop screens, phone displays, and desktop screens to view and interact with the data [2, 13]. In recent years, astrophysicists have started moving beyond 2D visualizations, finding other methods for effectively communicating their discoveries to journal readers. Two such examples include Augmented Reality (AR) tangibles and AR tabletop displays [1, 6, 17]. AR options have become a means for readers to explore 3D visualizations without the need to shift away from the context of the journal article (such as having to open a dedicated web-based 3D visualization [11]). These options also allow for authors to need far less technical expertise to generate AR figures than to embed a 3D visualization in an electronic-format research paper [3]. Previous work by Adams et al. has demonstrated the effectiveness and engaging qualities of these technologies through the evaluation of cognitive load associated with solving tasks using curated K-12 educational astronomy visualizations in both AR and non-AR environments [1]. Although the effectiveness and great potential of the technology were demonstrated, are we confident that these results hold for astronomers engaging with real astronomical data prepared for real publications? How confident are we that the results of this original study hold if different software is utilized? To answer these questions, we conducted a replication study.

A replication study is a technique to validate and expand the results of a previously executed study by re-running the study with either the original or adapted protocols [14, 36]. Replications within the academic community are essential to ensure that the prior results hold true [19, 20]. This is especially true in the field of data visualization, given how

young and rapidly developing the field is, where numerous theories have gone without validation through replication [18]. This is a component that some fields in social sciences deem necessary to consider a study complete [28]. The visualization community itself also recognizes the need for replications [35]. However, the current culture devalues replication studies, which are novel contributions that are vital to the field [18, 19, 33]. To ensure the theory and guidelines in our field are true and future work is valid, it is essential to verify past results [34].

We conducted a conceptual replication study, as illustrated in Fig. 1, in which the original study protocol was followed but with alterations to the data stimuli and software to expand the results and validity of the work. Like the original, our replication was a mixed-methods evaluation to assess the workload and feasibility of conventional 2D-on-phone versus two AR techniques to display 3D visualizations of astronomical data. The study was conducted at the same conference series, the 245th Meeting of the American Astronomical Society (AAS), two years after the original, with 34 participants in a nearly identical physical set-up in the exhibit hall to the original study.

This paper contributes: (1) The results of a conceptual replication study which validates the trends found in the original study and further validates the effectiveness and engagement of AR for 3D visualizations of astronomical data, and (2) a discussion and reflection for visualization researchers on the importance of fully documenting and disseminating materials to execute replication studies in meaningful ways.

2 RELATED WORKS

Augmented Reality (AR) as a technology has been used in a wide variety of settings, ranging from entertainment [23] to education [31] to industry [8, 26], to visualize information in a manner that projects the virtual environment to a real-world setting on a computing device. For AR, physical devices are sometimes used in tandem to display information and interact with the underlying visualizations, with the Merge Cube [24] being an example of such a physical device. These physical devices are known as “tangibles” and take many forms, such as physical markers, tags, cards, and more that can be identified by AR technologies [7]. One prior study has observed that AR with data visualizations has previously been outperformed by a desktop 2D display in precision, speed, and familiarity [2]. Other studies have focused on the interaction methods in AR, such as the use of tangibles [27]. In Adams et al. (2023) and our replication study, we go beyond prior work by evaluating three technology conditions (Non-AR on smartphone, Tangible AR, and Tabletop AR) based on workload experienced in each condition to better understand how users interact with visualizations with the AR and non-AR technologies through both quantitative and qualitative data analyses [1].

Replication studies are essential to ensure that prior findings hold true and that results are transferable and applicable to future studies [15] while being seen as a core of scientific procedures to solidify a foundation of knowledge to build from [20]. Despite the overall contributions to a field that replication studies provide, “replication crises,” in which work is not replicated, have broken out in fields including physics [4], psychology [29], and HCI [10, 16, 36]. Data visualization is beginning to follow down the same path in which replications are not valued in the same way that novel research is because the current publication culture devalues replications as being “unoriginal” and “not contributing novel results,” and, as such, trend toward being more cumbersome to publish [18, 19, 37, 34]. As we seek to avoid a similar crisis in data visualization, we follow these motivations to validate existing theory and findings through replication studies and conducted the conceptual replication study presented in this paper to validate and better understand the results of Adams et al. [1].

3 EXPERIMENTAL DESIGN

Study Design & Protocol: The conceptual replication followed the same mixed-methods (2x2 within-between subjects) study design with both quantitative and qualitative measures as in the original study. The within-subject factor was the comparison of Non-AR (on smartphone) versus AR (Tangible and Tabletop), and a between-subject factor of AR technique (Tangible versus Tabletop). The four parts of the study (with Part 1 repeated for two separate conditions), as shown in Fig. 1, are: **Part 0:** Before starting the study, participants were provided with a participant

information sheet and asked to digitally accept a consent form; **Part 1:** structured questions with tasks for participants to complete accompanied with the NASA-TLX questionnaire following the completion of tasks to understand workload experienced [12]; **Part 2:** Completion of the Mini-VLAT, a short test containing 12 questions on various tasks one needs to solve using various charts and figures [30]; **Part 3:** A survey to gather information about the participant such as their demographic information, familiarity with AR technologies, and information about their astronomy expertise; and **Part 4:** A period of open-feedback from the participants. The study length ranged from 16.5 minutes to 40.5 minutes, depending on the interest level and expertise in the use of AR technologies of the participant. Following the completion of the study, participants were compensated with a Merge Cube [24], with a retail value of \$29.99. This study was approved and executed in accordance with Northeastern University’s Institutional Review Board. The study protocol, questions, stimuli, and link to the preregistration are provided within the Supplemental Materials and in the following OSF repository: <https://osf.io/j8urq/>.

Conceptual replication differences: We chose to move the demographic survey at the end of the study, after completion of the task questions, to negate any impact the questions may have on biasing the study participants. We also added the Mini-VLAT section of the study in order to understand if a participant’s graphical comprehension related to their workload when solving tasks with 3D astronomy visualizations in our study [30]. As a consequence of this additional section in the study, the overall study duration was increased on average by 4 minutes.

Participants: The participants, as in the original study, consisted of 34 conference attendees who were part of the astronomy and astrophysics research community. None of the participants were a part of the original study by Adams et al. (2023). The study, conducted at the same conference series as the original experiment, took place from January 12th to 16th in the exhibit hall of the 245th AAS Meeting, 2025. The study results of one participant were removed from analysis due to technical difficulties, resulting in incomplete data collection. The population of the remaining participants included 16 men, 16 women, and one gender-diverse participant.

Conceptual replication differences: Great effort was put into replicating the study environment and participant pool of the original study; however, the study did take place in a different convention center. The participant demographics are close but not identical to the original, and we were unable to recruit as many participants primarily due to the longer study duration from the addition of the Mini-VLAT, more participant engagement, and longer discussions in Part 4.

Tasks & Stimuli: We used the same Latin squares ordering to evenly split participants in the AR condition and condition ordering, as well as task ordering. Participants solved six tasks that involved one or more of the same interaction types as the original study: panning, rotating, and zooming. Like the original study, these tasks were based on the ways users navigate 3D visualizations [5] and common astronomy visualizations [21]. The datasets encompass both 3D surface visualizations and 3D scatter plots [21], are suited for interactive tasks, and cover a variety of fields within astronomy. The NASA-TLX workload scale measured the task workload participants experienced [12]. For the qualitative analysis of the study (observations in Part 1 and open feedback in Part 4), we employed grounded theory with thematic analysis and axial coding [32]. Part 1 of the study was performed fully on a mobile phone, the same phone used in the original experiment, with the CoSpaces app for both the AR and Non-AR conditions [9].

Conceptual replication differences: In contrast to the original, we utilized *real astronomy data* from 3D AR figures in prep for or published in astrophysical journal papers. All study stimuli and tasks are provided in the Supplemental Materials, and one example is highlighted in Fig. 1. As real examples of astronomy data in 3D AR figures generated for publications were not available at the time of the original study, the evaluation used curated K-12 educational astronomy visualizations inspired by real data as stimuli in the Merge Explorer application [25]. Since the validation of the technology in Adams et al. (2023), the publication of 3D AR visualizations has expanded, providing stimuli for this work. The stimuli datasets were chosen to as closely as possible reflect the same visual appearance and subfields of astronomy as the original study (Fig. 1), but are not identical. In the replication, we utilized the CoSpaces application [9] (now called Delightex at the time of the

composition of this paper) as it is used by astronomers and enables a user to upload their data and author their own immersive visualizations. **Hypotheses:** The hypotheses of the study are, **H1:** AR will require a lower workload than non-AR technology, **H2:** Tabletop AR and Tangible AR will perform equally well in terms of workload, and **H3:** There will be no correlation in TLX scores with respect to the scores of the Mini-VLAT across participants. Given the findings of the previous study [1], as well as prior research based on performance involving spatial tasks across gender [22], we control for gender in our first two hypotheses.

Conceptual replication differences: The first two hypotheses are the same as the original study, but for the conceptual replication, we added the third hypothesis to expand upon the theory laid out by Adams et al. on how a participant's aptitude with certain measures may influence their workload under specific conditions. We opted to better understand a participant's aptitude to read visualizations and whether their ability to do so had any influence on their ability to reason with these mental models.

4 RESULTS

(H1) There was no significant difference in workload between the Non-AR Control condition and AR conditions, regardless of gender. As was done in the original study, we planned to conduct paired-sample t-tests to compare the self-reported TLX scores [12] for control (i.e., Non-AR on smartphone) and AR technologies (i.e., Tabletop and Tangible) [1]. One participant identified as gender-diverse and was excluded from the gender-based statistical analysis. The test for normality (Shapiro-Wilk) assumption for men failed ($p=0.01$) but succeeded for women ($p=0.56$). In light of this, we used the Mann-Whitney test for the men's group. Though not explicitly stated in the original study, it used a Bonferroni correction to lower the alpha necessary for significance due to investigating men and women separately, something we use again here and set $\alpha=0.025$ as our threshold for significance.

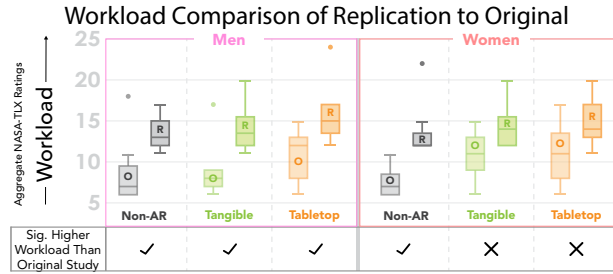


Figure 2: Quantitative NASA-TLX aggregate measures showing differences between and within the Replication (R) and Original (O) studies. The trends between gender and conditions were similar to the original; however, the reported workload was higher across all conditions for the men's and women's groups in the replication study, with all conditions for men being statistically significantly different from the original.

As shown in Fig. 2, there is no statistically significant difference observed between the workload in the Non-AR condition and AR conditions among women ($t(15) = -1.91, p=0.07$) (in contrast to the original study that was significant), nor men ($U=94.5, p=0.21$) (like the original study). While we do not derive statistically significant differences necessary to reject the null hypothesis, we observe a trend that for both men's and women's groups the average paired difference in TLX scores was lower for the Non-AR condition ($M=-1.31$ and $M=-1.56$, respectively).

Utilizing the measurements of H1, we conducted a comparison analysis of the original and replication study data. Specifically, we used data from the original study available via OSF to validate whether the increased workload (experienced by all participants across all conditions) was significant. We utilized the Mann-Whitney test for conditions that failed the Shapiro-Wilk test for normality, and two-sided t-tests for those that did not, with Bonferroni correction. We hypothesized that for men and women, there would be no difference between the original study's TLX scores and the replication study's TLX scores. Given a total of 6 tests to compare the scores - two for each gender analyzed for the

conditions: Non-AR Control, Tangible AR, and Tabletop AR - we set $\alpha=0.00833$ as our threshold for significance.

We found a statistically significant difference between studies in the following: Control condition for men (MW - $U=19.0, p<0.0001$); Control condition for women (MW - $U=5.0, p<0.0001$); Tangible condition for men (MW - $U=6.5, p=0.001$); and Tabletop condition for men (TT - $t(13.1)=-3.13, p=0.0079$). We did not find any statistically significant difference between studies in the following: Tangible AR condition for women (MW - $U=16.0, p=0.0114$) and Tabletop AR condition for women (TT - $t(15.3)=-2.65, p=0.0177$).

In summary, as shown in Fig. 2, there is a statistically significant difference between the original and replication studies for all conditions for men, while women only have a statistically significant difference in the Non-AR condition between studies. The average workload differential between Non-AR and AR conditions, however, is minor for men; women, on the contrary, have an appreciable shift in the average workload differential between Non-AR and AR ratings. Specifically, women experienced a significantly lower workload in the Non-AR condition in the original study relative to the AR conditions, resulting in a greater differential between conditions, compared to the replication study, which has a lesser differential between Non-AR and AR conditions.

(H2) There was no significant difference in workload between Tabletop AR and Tangible AR. We planned to conduct an independent samples t-test to compare TLX scores regarding AR technology, as in the analysis plan for the original study. The tests for normality (Shapiro-Wilk) did not pass for women, while homoscedasticity (Levene test) did pass for women (Tangible AR $p=0.04$ and Tabletop AR $p=0.76$; homoscedasticity, $p=1.0$). Both tests did pass for men (Tangible AR $p=0.18$ and Tabletop AR $p=0.09$; homoscedasticity $p=0.68$). Given that the test for normality failed for women but succeeded for men, we utilized the parametric independent samples t-test for the men's group and the Mann-Whitney test for the women's group. Like the original study, we failed to reject the null hypothesis for both men's ($t(14) = -1.03, p=0.321$) and women's ($U=27.5, p=0.707$) groups. Although the original study also failed to reject the null hypothesis, on average, men have a lower workload associated with the Tangible AR condition relative to the Tabletop AR condition. In our replication we see similar results, with men experiencing a lower workload on average when using the Tangible condition relative to Tabletop AR ($\bar{x}_{Tangible,M}=14.125, \bar{x}_{Tabletop,M}=15.875$), while women experienced a very similar workload between the Tangible AR and Tabletop AR conditions ($\bar{x}_{Tangible,M}=14.500, \bar{x}_{Tabletop,M}=14.857$).

Qualitative observations illuminate, in the replication, many of the same challenges experienced in the original study. Many participants who experienced the Tabletop AR condition were again reluctant to stand up and move around, occasionally requiring prompting from testers to move. Many of those same participants attempted instead to use interaction methods on the phone screen, in hopes that they could solve the task from a seated position. The Tangible AR condition with the Merge Cube [24], however, suffered several flickers in the projected visualization (moments when the projection disappeared and reappeared after some time), and users experienced difficulty with the software recognizing the tangible, both in the period when we were actively attempting to recruit participants by showing them the technology and while participants were engaged in the Tangible AR condition. We did not, however, have participants complain about the bimanual usage of the phone and cube simultaneously, as was prevalent in the first study. Additionally, no participants dropped the cube or the phone device during our replication study.

(H3) There was no statistically significant correlation in TLX scores with respect to the scores of the Mini-VLAT across participants. To test if the TLX scores were influenced by the scores of the Mini-VLAT across participants, we performed an Ordinary Least Squares (OLS) regression, utilizing the Mini-VLAT score participants received as an independent variable that could predict the TLX scores as a dependent variable [30]. We tested the reported workload received across each of the conditions against the Mini-VLAT scores in our study. The fitted OLS regression models for the Non-AR Control, Tangible AR, and Tabletop AR conditions, respectively, are: TLX Score = $11.0025 + 0.324 * (\text{Mini-VLAT score})$, TLX Score = $11.3388 + 0.3868 * (\text{Mini-VLAT score})$, and TLX Score = $15.7778 - 0.0556 * (\text{Mini-VLAT Score})$. None of the regressions had

statistical significance - Control ($R^2 = 0.05, F(1, 31) = 1.64, p = 0.21$), Tangible ($R^2 = 0.05, F(1, 16) = 0.84, p = 0.37$), and Tabletop ($R^2 = 0.00, F(1, 13) = 0.01, p = 0.93$). It was found that the Mini-VLAT score across all conditions did not significantly predict the TLX score of participants - Control ($\beta = 0.3243, p = 0.21$), Tangible ($\beta = 0.3868, p = 0.37$), nor Tabletop ($\beta = -0.0556, p = 0.93$).

We fail to reject the null hypothesis across all conditions that there is no correlation in TLX scores with respect to the scores of the Mini-VLAT across participants. Qualitatively, many participants did not care for the Mini-VLAT test while taking it, with several participants getting visibly frustrated or making remarks about the lack of intuitiveness of the test. This is consistent with findings in another study replicating the Mini-VLAT and receiving feedback from domain experts [38].

In addition to our replication and validation of the original work by Adams et al. with our first two hypotheses, we aimed to expand upon their theory of how a participant's aptitude with certain measures may influence their workload under specific conditions. We opted to include the Mini-VLAT as a measure of a participant's graphical comprehension, or their ability to understand visual representations of data. Ideally, we would have preferred to include a measure of a participant's aptitude with spatial reasoning tasks; however, we were unable to find any such tests that would not have greatly extended the study duration (e.g., more than doubling the original study's length with one test's addition). This would have been particularly problematic, as including a smaller test that increased the average time per participant by approximately 4 minutes still resulted in us seeing approximately 35% fewer total participants relative to the original study. Due to the nature of the venue and our recruitment process, our statistical power would have further decreased if we had included a measure for spatial reasoning, and as such, it was not feasible to include one, despite its higher perceived relevance to the nature of the technology and data.

Qualitative Analysis: Based on our thematic coding of the participant-provided feedback (Sec. 3), three key themes were identified, with the final theme comprised of two sub-themes. The first theme is *Software Difficulties* with the application derived from either *reluctance to stand* during tabletop conditions, *Visualization Projection Flickering* when moving the Merge Cube [24] around, or *Unintuitive Interactions* during the Non-AR Control condition. Following the study's qualitative feedback, we performed several application tests comparing Merge Explorer [25] to CoSpaces [9] for the Tangible AR condition, and found that there were notable differences in the tracking of the Merge Cube that could have led to a higher workload experienced and negative qualitative feedback for the Tangible AR condition in our replication study, relative to the original study. For additional information on testing we performed that validated user experiences with the Tangible AR condition, please refer to Supplemental Materials.

The next theme is *Accessibility*, which arose in part due to some of the software difficulties. *Far-sightedness* impacted the ability of participants with that vision impairment to adequately use the Tangible AR condition in certain circumstances, due to the zoom breaking in certain arrangements of the cube relative to the phone's camera. The experience was certainly less optimal in some cases than that of participants who did not need corrective lenses, with some participants acknowledging that they did not bring their glasses, or having verbal regrets, such as "I should have my reading glasses on." *Physical differences*, such as tremors, led to the visual projection of the stimuli onto the cube being lost in some instances, with one participant reacting, "Whoops, where did it go?" Despite the difficulty with the cube, that same participant said the experience was "Amazing still." At the recommendations of the previous study, we had a phone stand ready to go, such that participants with physical differences did not have to hold both the phone and the Merge Cube at the same time, known as the "juggling" problem, which was greatly appreciated.

Despite difficulties, there was a great deal of positive *Engagement* with both the technology and the stimulus data. Broadly, many participants simply said "Wow" or similar utterances. For *technology-based engagement*, one participant said "It's like playing video games" when utilizing the AR Tabletop technology, while other participants simply said "This is fun" when using AR broadly, or were extremely interested in applying the AR technology to their data. During the open-feedback portion of the study, we allowed participants to try the other AR condition,

so that ultimately, all participants had some degree of interaction with the Merge Cube technology. Approximately half of the participants had comments that were similar to what one participant said: "It's easier, it's more tactile, and it's like magic." For *data-based engagement*, many participants were enthralled with the data, and some participants wanted to *keep exploring* beyond the scope of the task they were asked to complete. We found that participants who were particularly engaged with the data either took *less time solving* or more "*easily*" *solved tasks* compared to other participants. The "wow factor" was certainly more prominent across participants who were more engaged with the data, with several participants simply in awe of the data that was visualized.

5 DISCUSSION & FUTURE WORK

A key takeaway from the results of the original study was the observed *tradeoff between workload and engagement*. The results of our replication confirm this and provide additional illuminating evidence of the tradeoff. As discussed in Sec. 4, the replication measured an even higher workload, but also, through the qualitative evidence, even higher engagement. We believe the increased workload was in part due to the technical difficulties encountered and the higher level nature of the *real astronomy data*. However, the real astronomy data also led to an increase in engagement of participants as compared to the educational astronomy visualizations in the original study. In the replication, the stimuli resulted in more time spent in Part 1 as participants either were curious about new astronomical discoveries or excited to explore a concept or object familiar to their work. This was especially evident in Part 4 when participants took more time to express their thoughts on their experiences. In spite of the additional workload required for AR overall, the participants continue to be very excited, positive, and engaged about the use of AR for presentation in scholarly articles to better communicate their own astronomical data discoveries. Many of the study participants were very intrigued by the idea of using AR for 3D visualizations as part of their research and data analysis process and looked forward to exploring this potential in future work.

We highlight the *importance of replications* in our study, and of understanding subtle and major differences between the studies. In our study, software differences were a subtle example as CoSpaces [9] operated fundamentally the same as the Merge Explorer [25] as an AR viewer, albeit with CoSpaces having a slightly worse tangible tracking algorithm (it should be noted that in other studies, different software may contribute a larger difference in participant experiences). A major difference in our study, however, came through in the form of stimulus differences, as our new stimuli required a higher level of understanding to solve tasks. In our replication, other differences that we did not measure (and were not measured nor reported in the original Adams et al. study) but could have subtly impacted the study outcomes include hardware degradation, environmental lighting, and conference location impact on demographics. We include a full list of differences we hypothesize, consequential or not, in our Supplemental Materials to make this work replicable in the future and bolster the transparency of our findings. We believe that how we execute our replication study and report the differences between our study and the original study serves as a meaningful guideline for future researchers seeking to validate prior findings in other work. It is to provide sufficient information to perform a replication, even details that an experimenter may believe are not important or consequential. Through replications, our visualization research community can validate theory and gain deeper insight, strengthening past work and informing future work.

6 CONCLUSION

We replicated Adams et al.'s evaluation of Non-AR Control, Tabletop AR, and Tangible AR with real astronomy data as opposed to curated educational data [1]. The results of our replication highlight an overall increase in the workload experienced by participants relative to the original study. We find that the technologies evaluated are all still viable options for visualizing astronomical data, with a decrease in the tradeoffs mentioned in the previous study between workload (where AR required more physical and mental work) and engagement (where AR was more engaging to participants). Given the impacts of changing the stimuli and the technology on participant workload experiences, we encourage the data visualization community to continue the trend of replicating prior work to ensure that results found on representative data extrapolate to real-world data in applied settings.

ACKNOWLEDGMENTS

The authors wish to thank the entire *glue* team and Jane Adams for their helpful input on this research and paper. We also thank the AAS for their support and collaboration. This work was supported in part by the National Science Foundation (#2209623, #2209624, #2209625) and the Khoury College of Computer Sciences, Northeastern University.

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